

CASE STUDY

AI-Powered Integrated Operations Platform (IOP)

For a large manufacturing company seeking real-time supply chain visibility, forecasting, and connected operations

Industry	Scope	Outcome
Manufacturing / Supply Chain	Planning, logistics, quality, reporting	One connected command center

Challenge

The client relied heavily on legacy Excel sheets and manual coordination across departments, which made it difficult to keep data current, consistent, and actionable.

Teams lacked a live view of sales, procurement, logistics, inventory, and quality activities, causing delays in planning and response.

They also needed a system that could integrate with their existing operational ecosystem, including Syspro, while giving leadership a faster way to monitor performance and make decisions.

Solution

- We developed a customized AI-integrated supply chain platform with a central command center that unified SKU information, alerts, and live operational updates.
- Forecasting modules were built to support sales planning, raw material procurement, top SKU analysis, and revenue visibility, helping teams plan ahead with confidence.
- A logistics layer brought truck tracking, shipment visibility, on-water monitoring, arrival management, booking management, and PO tracking into one operational view.
- A quality management workflow was added to improve shipment control, reduce scrap, and connect quality insights back into the wider supply chain process.
- All departments were connected on a single live system so every team could work from the same source of truth and stay ahead of expected events using AI-driven predictions.

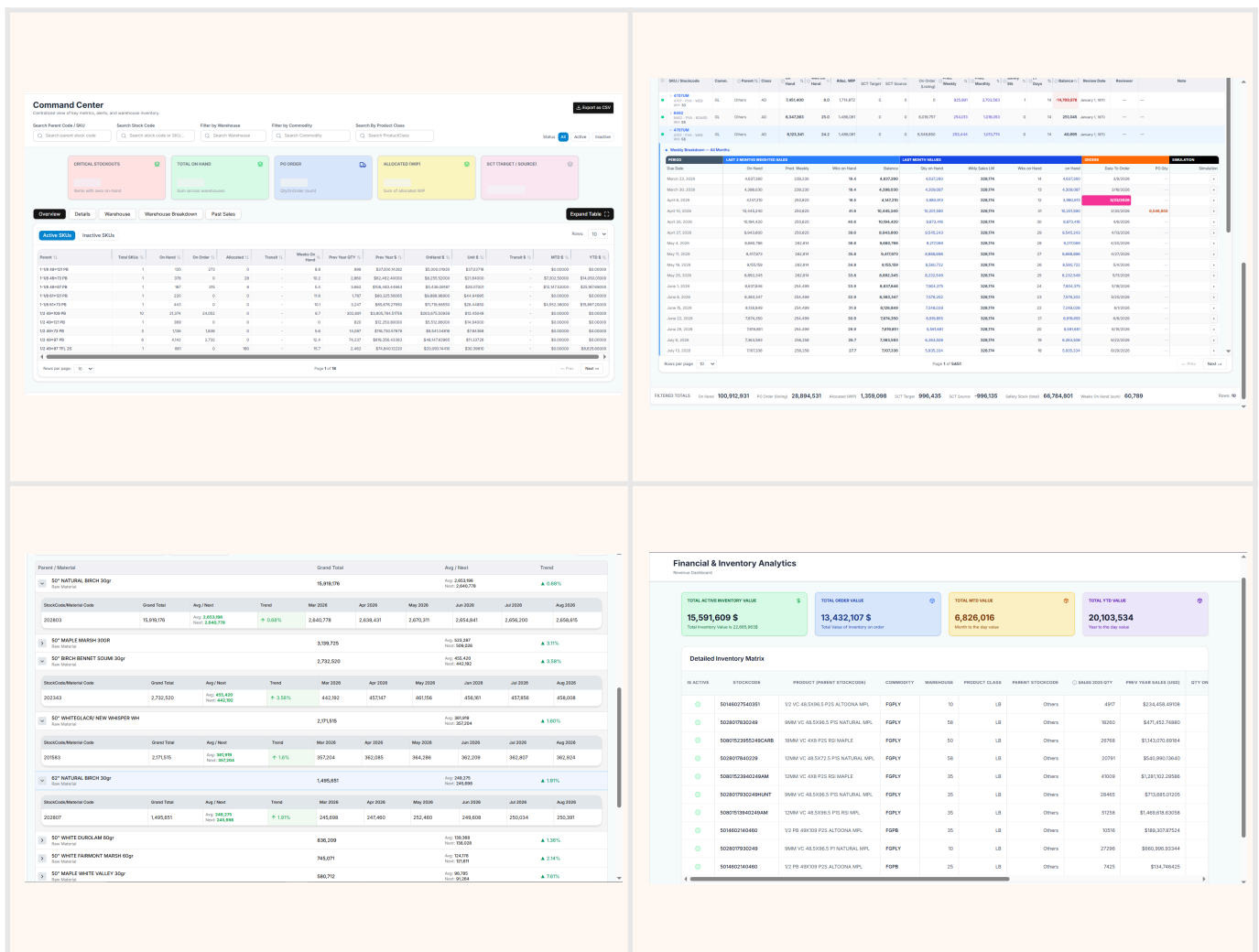
Impact

Operational Clarity Replaced static spreadsheets with live dashboards and a centralized command center.	Faster Planning Improved visibility for sales forecasting, raw material needs, and revenue planning.
Better Logistics Control Enabled end-to-end monitoring of shipments, trucks, bookings, and PO movement.	Quality Improvement Supported proactive quality management and helped reduce scrap through connected data.

Technology Stack

Layer	Details
Frontend	React.js / Next.js based dashboards and command center views
Backend	API-driven application architecture for data flow and operations
Data Layer	Scalable SQL / NoSQL storage for live business records
AI / Forecasting	Predictive models for sales, raw material, and performance insights
Integration	Syspro and other operational systems
Infrastructure	Cloud-hosted, scalable, and secure deployment environment

Screenshots:



Final Notes:

By replacing disconnected spreadsheets with a customized AI-powered integrated platform, the client gained a single operational view across supply chain, logistics, quality, and planning. The result was a more

connected, proactive, and scalable operating model designed to keep every department aligned and ahead of what is coming next.